EE Dept, IIT Bombay

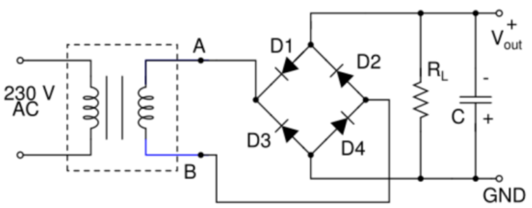
Academic Year: 2025-2026, Semester I (Autumn)

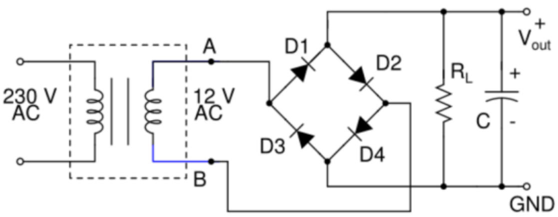
Course: MS101 Makerspace

EE Lecture 5: Diode Circuits

Sample Problems

1. Two wrong bridge rectifier circuits with a capacitor filter are given below. Rectify the mistakes and draw them correctly.

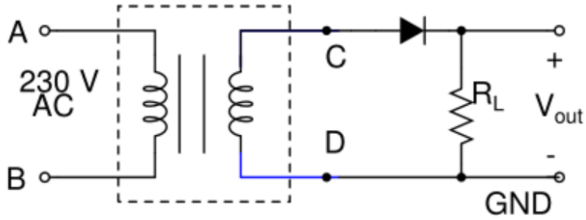




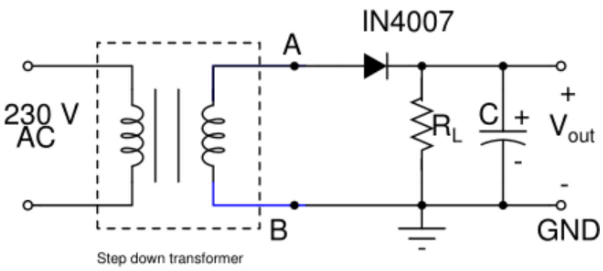
2. The half-wave rectifier circuit shown below, uses a 230 V(RMS) to 12 V(RMS) step-down transformer.

A) If resistor R_L is removed from the circuit, what will be V_{out} in volts?

B) If the diode is now reversed, sketch V_{out} .



3. The rectifier circuit shown below uses a 230 V (RMS) to 15 V (RMS) step-down transformer.



A) Evaluate the average value of V_{out} if C is removed.

B) With the capacitor C present and the load resistor R_{L} not present or very large, evaluate V_{out}.

C) Find the V_{out} as in (B) if the transformer input voltage is 115 V (RMS).

4. Which of the following unregulated DC power supplies, all having the same output DC voltage (average value of V_{out}), has the highest peak-to-peak ripple voltage?

A) An unregulated DC power supply made of a half-wave rectifier with load resistance R_L = 2 kΩ and a filter capacitance C = 100 µF.

B) An unregulated DC power supply made of a half-wave rectifier with load resistance R_L = 2 kΩ and a filter capacitance C = 1000 µF.

C) An unregulated DC power supply made of a bridge rectifier with load resistance R_L = 2 kΩ and a filter capacitance C = 100 µF.

D) An unregulated DC power supply made of a bridge rectifier with load resistance R_L = 2 kΩ and a filter capacitance C = 1000 µF.

5. Two rectifier circuits are shown below. With reference to these circuits **mark all statements that are correct.**

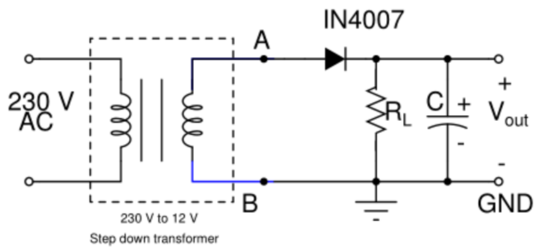


Fig 5A

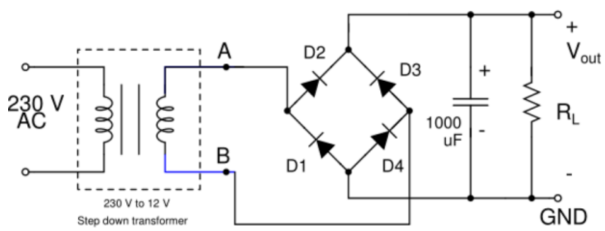


Fig 5B

A) For the same values of R_L and C, the average value of V_{out} of the circuit in Fig 5A will be greater than that in Fig. 5B.

B) For the same values of R_L and C, the average value of V_{out} of the circuit in Fig 5B will be less than that in Fig. 5A.

C) For the same values of R_L and C, the average values of V_{out} of the circuits in Fig 5A and Fig 5B will be the same.

D) For the same values of R_L and C, the peak-to-peak ripple voltage in V_{out} of the circuit in Fig 5A will be greater than that in Fig 5B.

E) For the same values of R_L and C, the peak-to-peak ripple voltage in V_{out} of the circuit in Fig 5B will be greater than that in Fig 5A.

F) For the same values of R_L and C, the peak-to-peak ripple voltages in V_{out} of the circuits in Fig 5A and Fig 5B will be the same.

6. The circuit diagram for a clipping circuit for peak limiting is shown below in Fig P6. Given: R = 2 kΩ, diode forward voltage drop = 0.7 V, v_I = 6 sin ωt.

A) What is the maximum value of the diode current in mA?
B) What are the minimum and maximum values of v_O in volts?
C) Sketch v_I and superimpose on it the diode current and the v_O waveform.

7. The circuit diagram for a clipping circuit for valley limiting is shown below in Fig P7. Given: $R = 2$ kΩ, diode forward voltage drop = 0.7 V, $v_I = 6 \sin \omega t$.

A) What is the maximum value of the diode current in mA?
B) What are the minimum and maximum values of v_O in volts?
C) Sketch v_I and superimpose on it the diode current and the v_O waveform.

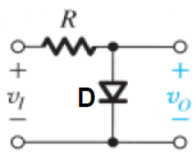


Fig P6

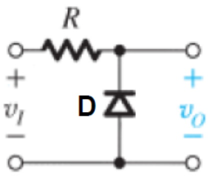


Fig P7

8. The circuit diagram for a clipping circuit for peak and valley limiting is shown below in Fig P8. Given: $R = 2$ kΩ, diode forward voltage drop = 0.7 V, $v_I = 6 \sin \omega t$.

A) What are the minimum and maximum values of the diode current in mA?
B) What are the minimum and maximum values of v_O in volts?
C) Sketch v_I and superimpose on it the two diode currents and the v_O waveform.

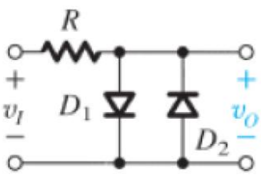
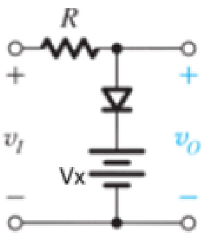


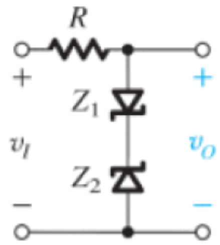
Fig P8

9. The circuit diagram for a clipping circuit for peak limiting with a voltage source is shown below in Fig P9. Given: $R = 2$ kΩ, Vx = 2 V, diode forward voltage drop = 0.7 V, $v_I = 6 \sin \omega t$.



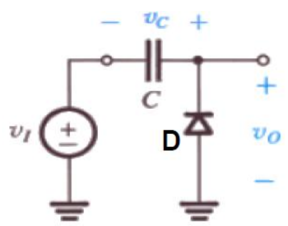
A) What is the maximum value of the diode current in mA?
B) What are the minimum and maximum values of v_O in volts?
C) Sketch v_I and superimpose on it the diode current and the v_O waveform.

10. The circuit diagram for a clipping circuit using Zener diodes is shown below in Fig P10. Given: $R = 2$ kΩ, $V_{Z1} = V_{Z2} = 3$ V, $v_I = 8 \sin \omega t$. Assume that the Zener diodes have forward voltage drop of 0.7 V and vertical I-V characteristic in the breakdown region.



A) What are the minimum and maximum values of the current through the Zener diodes in mA?
B) What are minimum and maximum values of v_O in volts?
C) Sketch v_I and superimpose on it the diode current and the v_O waveform.

11. A clamping circuit is shown in Fig P11. Given: $v_I = 4 + 8 \sin \omega t$ V, diode *D* is ideal, i.e. it has zero forward voltage drop.



A) In the steady state, i.e. several cycles of v_I, what will be the value of v_c in volts?
B) What will be the peak value of v_O in volts?
C) What will be the minimum voltage of v_O in volts?
D) Sketch v_I and superimpose on it the v_O waveform.

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